| INCH-POUND |

MIL-C-48171(AR)
AMENDMENT 7
20 May 1990
SUPERSEDING
AMENDMENT 6
05 July 1988

MILITARY SPECIFICATION

CARTRIDGE, 81MM, HE, M374A3
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-C-48171 (PA), dated 14 January 1975, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, To the list of Military Specifications, add the following:

"MIL-C-48156 - Cartridge, Ignition, M299, Loading, Assembling and Packing".

Following line 21: Add "9312648 - Cartridge, 81MM, HE, M374A3 W/O Fuze".

Line 22: Drawing "9241291" delete "W/Fuze, PD, M524A6" and substitute "W/Fuze, PD, M567".

Line 24: Delete "9230175 - Container ... in Jungle Wrap" and substitute "9276596 - Container Ammunition, Fiber, M252A5 for 81MM Mortar Ammunition".

At bottom of page: Add "9287603 - Packing and Marking for Container, Fiber, M252A5, Ammo for 81MM Ammo in Jungle Wrap".

PAGE 2

2.1, Add under Inspection Equipment Drawings as follows: "9280418 - Fin Distortion Gage".

3.2, Line 2: Add "or 9312648, whichever is applicable."

PAGE 3

3.5.2, Delete "function" and substitute "function high order".

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4.2.2, Line 4: Delete "Inspection for Critical Defects (and Major Defects, when so specified)." and substitute "Inspection for Critical Defects and Special Defects (and Major Defects, when so specified."

Line 12: Examination, insert the following before the last sentence of the paragraph:

"The contractor may elect to perform 100% inspection with automated equipment or impose mandatory process controls on automated operations in lieu of conducting sampling inspection for defects listed in this specification. In such cases, calibration plans shall be submitted to the Technical Agency (see 6.8) for approval for each automatic inspection station and process control plans must be submitted for approval for each of the automated operations. The plans shall identify; (1) the frequency (production interval) for validating equipment functioning, (2) the defect standards and (3) a description of material control procedures to be imposed on product passing through each automated station during the production interval between validations. Under these provisions final acceptance will be withheld from material produced during a production interval until satisfactory validation of each of the automated stations so controlled. In the event of a validation failure of any station, the material processed through such station since the last acceptable validation shall be placed in a rejected status pending corrective action."

PAGE 6

4.2.2.1, Line 2: Delete "9241291)" at end of line and add "9241291 or 9312648."

4.2.2.2, Line 3: Delete "9241291" and add 9241291 or 9312648."

4.2.2.2, Line 8: Delete "101. Depth ... max ... Gage CD" add new catagory heading, "Major: 100 percent inspection 101. Depth of fuze well, max ... Gage CD."

4.2.2.3, Line 2: Delete "9241291" and add "9241291 or 9312648."

Line 6: Delete "Critical 2. High explosive extends beyond top of liner ... Visual" in its entirety.

4.2.2.3.1, Add new paragraph as follows:

"4.2.2.3.1 Charge, Propelling, M205 (see dwg. 9280588), Note a

Critical 1. Container or container seal damaged to the extent that propellant shall escape ... Visual (see 4.3.4)

Major 101. Container or container seal damage but not to the extent that propellant shall escape (100% inspection) ... Visual (see 4.3.4)

Minor: None defined

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Note a - If LAP of the propelling charge, M205, and the cartridge, M374A3, are performed at the same installation, this paragraph is not required."

4.2.2.4, Line 1: Delete in its entirety and add "4.2.2.4 Cartridge (see dwg. 9241291 or 9312648)"

Line 3: Delete "Critical 1. Ignition cartridge head not fully seated . . . Visual" and add "Critical 1. Depth of ignition cartridge from fin end below minimum ... Gage CD"

PAGE 7

4.2.2.4, Line 1: from top of page: Delete line in its entirety and add "2. Fuze safety wire missing or insecure (if applicable; see dwg. 9241291) ... Visual-Manual"

Delete critical 3 and add new critical 3 as follows:

"3. Propell ing charge missing or damaged to the extent that propellant shall escape (if applicable, see 4.3.4.1) ... 100% ... 3.2 ...Visual (see 4.3.4)"

* Line 5 from top of page; Delete line in its entirety and add "6 ... Cracks in fin housing ... Visual (4 power magnification)"

* Following line 5 from top of page: Add: "Special a: Warning notice(s) missing or illegible ... Visual (see 6.7)"

Line 10: Delete line "103. Outside diameter, max ... Gage CD"

* Add new Major 105 as follows:

"105 Fin blade damaged or distorted ...100% inspection...3.2...Gage (see drawing 9280418)"

Line 20 from top of page: Delete line in its entirety and add "108. Fuze fails to meet mnimum assembly torque (if applicable; see dwg. 9241291). Test"

Line 24 from top of page; Delete line in its entirety and add "111. Fuze not fully seated (if applicable; see dwg. 9241291 ... Visual"

Add new 112 as follows:

"112. Propelling charges damaged but not to the extent that propellant * shall escape (if applicable, see 4.3.4.1) (100% inspection) 3.2 ... Visual (see 4.3.4)"

Line 28 from top of page: Delete "Minor: 202. Pettman cement does not cover ... Visual" in its entirety and substitute "Minor: 202. Pettman cement (or alternate) improperly applied (applicable to the fuze, fin and ignition cartridge) ... Visual"

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Line 33 from top of page add "Minor 205 Closing plug not fully seated (if applicable; see dwg. 9312648) ... Visual"

4.2.2.5, Line 1: Delete "see dwg. 9230175" and substitute "see dwg. 9287603"

Line 3: Delete "Critical: None defined" and substitute Critical 1 and Special A. as follows:

"1. Propelling charge damaged to the extent that propellant shall escape 25(a) ... Visual (see 4.3.4)

Special A. Propelling charge support missing Visual"

Major catagory, delete "None defined" and substitute 101 as follows:

"101. Propelling charge damaged but not to the extent that propellant shall escape 25(a) ... Visual (see 4.3.4)"

Add note (a) as follows:

"(a) Twenty-five cartridges which have the support for the propelling charges assembled shall have the support removed and remove the four propelling charges. The propelling charges shall be inspected for damage (splits, cracks, tears) which may result from assembling/disassembling of charges and support on fin boom. The inspection plan for major defects shall be 100-1-2".

PAGE 8

4.2.2.6, Line 1: Delete "see dwg. 9230175" and substitute "see dwg. 9287603"

4.2.2.7, Line 1: Delete "see dwg. 9230175" and substitute "see dwg. 9287603"

Following line 10: Add: "203. Kraft paper missing or not properly assembled ... Visual"

Add the following paragraph:

"4.2.2.9 Liner (See dwg. 7549011).

Catagories	Defects	Method of Inspection
Major	100 percent inspection	Gage
101.	Length, min.	
Major		
102.	Length, max.	Gage
103.	Outside diameter, min.	Gage"

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Add the following paragraph:

"4.2.2.10 Packing and Marking for Box, Wirebound Packing, Ammunition for 81MM Mortar Ammunition in Jungle Wrap (see dwg. 9287601).

Catagories	Defects	Method of Inspection
Minor		
201.	Car seal missing, unsealed or improperly positioned	Visual"

Add the following paragraph:

"4.2.2.11 Support (see dwg. 9369933)

Catagories	Defects	Method of Inspection
Major		
101.	Inside length	Gage
102.	Rib radius	Gage
103.	Inner radius (small)	Gage
104.	Length between open end	Gage"

4.2.3.1, Line 2: Delete "XM299" and substitute "M299"

PAGE 9

4.2.3.2: Delete entire paragraph and substitute "Cavitation (see 3.3 - except 3.3.4) Major Defect - The sampling plan and procedure for this test shall be in accordance with MIL-STD-1235 using an AQL of 0.65 percent. Any cartridge having cavitation in excess of that permitted by the applicable requirement for any segment shall be classified defective. The test shall be performed as specified in 4.3.1."

4.2.3.4, sub-paragraph e: Delete "e. Metal parts separation, critical defect (see 3.5.5)." and substitute, "e. Metal parts separation, special defect (see 3.5.5)."

PAGE 10

4.2.3.4.2, Line 2: Delete "4.2.3.3.1" and add "4.2.3.4.1".

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PAGE 11

4.3.2, Line 5: Delete "one half of the sample cartridges shall be fired at charge one and the other half at charge nine." and add "One half of the sample cartridges shall be fired at charge zero and the other half at charge four".

Line 8: Delete "The individual velocities, pressures and ranges shall be recorded on the firing record" and substitute "The individual velocities, presures (charge four only), deflections and ranges shall be recorded on the firing record".

* Line 10: Delete "In the event of a misfire, the ignition cartridge shall be removed and forwarded to Picatinny Arsenal" and substitute "In the event of a misfire, the ignition cartridge shall be removed and forwarded to ARDEC, ATTN: SMCAR-AEE-BP".

Line 31: Delete "...and prints of all the other rounds in that particular charge zone, shall be attached to the firing record." and add "...and prints of all the other rounds in that particular charge zone, in that lot, shall be attached to the firing record."

PAGE 12

4.3.4, Add new paragraph as follows:

"4.3.4 Damaged propelling charge. To ascertain that a damaged propelling charge is to be classed as critical or major, the following procedure shall be used:

a. Place the damaged charge on a M24 fin, oriented such that the propellant would be resting against the damaged surface.

b. The fin-charge shall be hand-shaken vigorously three times.

c. Examination shall be made for propellant spillage - if any propellant is observed, the damaged propelling charge shall be classed as a critical defect if no propellant is observed, the damaged propelling charge shall be classed as a major defect.

4.3.4.1 Alternate assembly/inspection procedure. The contractor may elect to utilize the following assembly operations.

a. Place the support for the propelling charge in a holding fixture.

b. Properly place the four propelling charges in the support with gaps aligned upward.

c. Press the fin housing, into the propelling charges and support assembly (step b).

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d. 100% visual inspection for damage to the ends of four propelling charges will be performed.

e. Continue the assembly/inspection operations (see 4.2.2.5)."

5.2: Delete this paragraph in its entirety and substitute

"5.2 Packaging, packing and marking - Packaging, packing and marking shall be in accordance with dwg. 9287603, 9287601 9276596".

PAGE 13

6.2.2, Line 15: Delete "The Contractor if referred to MIL-HDBK-204, 'Inspection Equipment Design' for guidance."

Line 17: Delete "five digit" and substitute "paragraph and".

* 6.3, Line 6: Delete in its entirety and add "ATTN: AMSMC-QAT-B(D), SMCAR-FSA-IM(D)."

PAGE 14

Add the following paragraph:

"6.7 Special defect. A defect, other than Critical, that judgement and experience indicate may, depending upon the degree or variance from the design requirement:

a. Result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product or

b. Prevent performance of the tactical function of the major end item."

Add the following paragraph:

* " 6.8 Submission of calibration and process control plans. Prior to implementation of calibration or process control plans in accordance with 4.2.2, plans shall be approved by ARDEC, ATTN: AMSMC-QAT-I(D), Picatinny Arsenal, New Jersey 07806-5000."

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